

EC348: Assignment 1 - 5538981

1. Introduction:

This report looks at the findings from Chakravorty et al (2023), which examines the short-term and long-term effect of the COVID pandemic on rural unemployment amongst India's migrant workers. Their findings of the marginal effect of completing the DDU-DKY program on finding a post-pandemic job is replicated, examined, and linked to additional relevant literature. Finally, this report extends the original analysis by applying alternative model specifications, linking the findings with more recent literature, and using machine learning techniques to examine the most important variables in rural employment outcomes.

2. Literature Review:

Youth unemployment is an indicator of the wasted potential of a country's human capital, as the sufficient development of a country's youth is critical for the nation to develop sustainably in the future (Dev and Venkatanarayana, 2011). It has been an issue that has consistently plagued India, being recognised as a source of concern as far back as 1988 with the formulation of India's National Youth Policy. Despite this, both inaction and a lack of effective policies towards reducing youth unemployment have enabled the issue to persist (Visaria, 1998; Sharma, 2022). Literature finds that unemployment enters workers into a vicious cycle of reduced mental well-being and heightened stress, resulting in higher suicide rates and hindering the development of future human capital (Jones and Sen, 2022; Sinha, 2018). At a national level, this exacerbates poverty, malnutrition, and inequality (Nair, 2020; Dev and Venkatanarayana, 2011).

Whilst multiple studies examine the negative effects of COVID on unemployment, studies conducted during the pandemic face the common limitation of the short-term nature of the data that was available to them. Therefore, it is important to investigate the impact of a unique event such as COVID for a sufficient time after it has ended to examine its long term impacts and derive policy interventions based on the experiences of the youth (Blustein et al, 2020).

Furthermore, the disproportionate impact of COVID on marginalised communities warrants investigation to better inform policy makers regarding existing inequalities (Couch et al, 2020; Bauer and Weber, 2021; Blustein et al, 2020). The rural community is marginalised in India: as such, it is important to investigate how COVID impacted them. It should be noted that there is some debate

regarding the importance of rural unemployment to the wider Indian economy, as studies find it insignificant in inciting theft; additionally, the increasing prevalence of white-collar crimes threatens to deprioritise the issue of rural unemployment for policy makers as white-collar crimes have a larger financial impact (Nagasubramanian and Joseph, 2024; Hazra, 2020)

3. Data:

Chakravorty et al (2023) utilise a survey of workers who had previously been enrolled in the Deen Dayal Upadhyay Grameen Kaushal (DDU-DKY) program. The program targets youths in rural areas aged 15-35 years old from poor families, and provides a mix of classroom and on-the-job training (Chakravorty et al, 2021). The study begins with 2259 respondents and ends with 1955 owing to a 15% attrition rate. There were 3 survey rounds after the baseline survey, which questioned participants on characteristics (such as employment, location, willingness to migrate, and wellbeing) and employment-related outcomes (such as the number of applications they had made and whether they had obtained a job). Additionally, the participants were questioned on their use/knowledge of the Yuvasampark app to investigate the effectiveness of the platform in helping rural youths find jobs.

4. Empirical analysis:

4.1.0 Replicating Table 2:

Figure 1a shows the differential impact of completing training as part of the DDU-DKY program on the probability of salaried employment across all survey rounds. There is a consistent advantage of having completed training on the probability of being employed across all survey rounds. Unlike with the differential impact of gender, the regression is robust to the addition of control variables, indicating a causal relationship.

A limitation of this data is the nature of the outcome variable, which is a binary variable that takes the value of 1 if a participant possesses a job, and 0 otherwise. Underemployment has been a consistent substantial issue in India, where a lack of jobs has forced overqualified graduates to engage in low-paid jobs or working for fewer hours (Dev and Venkatanarayana, 2011). The pandemic only exacerbated the jobs shortage, increasing the pressure on trained workers to settle for lesser jobs. Hence, the variable only provides information on whether someone has obtained a job or not, and cannot provide insights

into the suitability of the job that the participant has received, or whether a trained participant was able to regain their pre-pandemic wage level faster than those who had dropped out of the training.

4.1.1. Extension - Interactive gender effect

This analysis can be extended by examining whether there is an additional effect of being treated for female participants compared to male participants. Figure 1b shows the relevant coefficient estimates for the treatment variable when including an interaction between the Training_complete and gender variables.

The table shows a significant effect (at the 1% level) of being trained for women in the first two survey rounds compared to men. This aligns with the finding of low female labour participation rates in India (Sharma 2022, Bairagya, 2015, Chahal and Rani, 2025). It indicates that being trained increased the competitiveness of rural women and increased their likelihood of being employed, showing the success of the DDU-DKY program on combating gender differences. However, this effect vanishes in the last two survey rounds, owing to how most male workers took up informal employment after the pandemic whilst most female workers dropped out of the labour force to take up domestic work (Chakravorty et al, 2023). This could be caused by the strong prevalence of social norms in rural states that relegate women to household work with the expectation of men to be breadwinners.

These findings therefore show that whilst programs such as DDU-DKY are successful in increasing the probability of formal employment for rural women, work still needs to be done to tackle social norms to encourage women to stay in the labour force even in times of uncertainty.

4.2.0 Replicating Table 4, Panel B:

Figure 2 shows the impact of the Yuvasampark intervention (where the treatment group was encouraged to sign up for the Yuvasampark app) on the probability of applying for a job in the last 2 months and the number of job applications made. The table shows that across survey rounds 2 and 3, there was an insignificant impact of the intervention, which the authors discuss as being caused by a lack of external infrastructure, a lack of jobs, a lengthy registration process, and language barriers (as the training modules were in English rather than the local language).

These findings align with literature that assert that adoption of modern technology is not a silver bullet (Jones and Sen, 2022; Kelley et al, 2021, Sharma, 2022). The Yuvasampark app lacked sufficient available opportunities for its participants, illustrating that a main obstacle in overcoming rural

unemployment is job creation rather than search frictions. The insufficient range of job postings and employers available on the app contributed to its infrequent use (Table A7, Chakravorty et al, 2023). Jones and Sen (2022) find that in a poor job market, the intervention of an online portal can either have no effect or be detrimental to participants' mental health owing to the fact that the portal visualises participants' failures to meet their job aspirations: in this experiment, this could have added to the decline in participants' mental health.

4.2.1. Extension – Selection of control variables

The authors use a double lasso selection method to choose control variables for this regression, where the selection resulted in no control variables being selected, illustrating that the control and treatment groups were sufficiently randomised since none of the variables were sufficiently related to both groups. This is reinforced by comparing the coefficients from Table 4 with the coefficients from a basic regression with only the outcome and treatment variables, and a regression with bluntly added control variables. There are no meaningful differences between the three sets of coefficients, indicating that the treatment is orthogonal to the control variables.

4.2.2 Extension – Random Forest model

I extend the analysis by Chakravorty et al (2023) by building a random forest model for the dataset to reveal which variables were most important in predicting the main employment outcomes of interest (employment status and number of job applications made). A random forest model is advantageous because of its non-parametric nature: it is better in adapting to non-linearities in the dataset than a standard regression model which imposes linearity restrictions ([R-project.org](https://www.R-project.org/), 2025).

Figure 2b shows a breakdown of the most important variables in the model's prediction of the outcomes of interest, showing that across all three outcomes, the most important predictors of obtaining salaried employment and the number of applications made were the Big 5 test scores (for Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness), owning one's own house, one's scores for the ASE Test, and one's scores for the Grit Test.

It is interesting that owning a house is a strong predictor for having salaried employment. Around half of the participants used informal channels to search for jobs as opposed to formal methods like online job portals (Figure A5, Chakravorty et al, 2023). In this context, having one's own house is an indicator of wealth and potentially the quality of connections that one has (as wealthier people may know other influential individuals) to provide them with paths to salaried employment. Whilst this finding shows

the importance of connections for finding employment in rural India, it is worthwhile to note that one's scores on the Big 5 and Grit tests are consistently strong predictors for both salaried employment and the number of applications, asserting that paths to employment are also predicted by ability and personality.

4.3.0 Replicating Table A9

Figure 3 shows a balance test for the treated and control groups on a number of observable variables, controlling for the strata effects of gender, sector of employment, state, and being in the treatment group. Only 5 variables show significant differences between the two groups: number of joint households and having 3(+) earning members in the family are significantly different at the 10% level; preferred annual earnings, having 3(+) relatives that migrated, and getting more than 50% on the Matric exam are significant at the 5% level.

4.3.1 Extension – Finding effect sizes

A simple t-test is not sufficient to determine whether these are meaningful differences between the two groups, as statistical significance can be influenced by sample size and it does not indicate the strength of the relationship between different variables (Tomczak and Tomczak, 2014; Sullivan and Fein, 2012). I therefore calculate effect size as a measure of practical significance using Cohen's d, where for each variable, I divide the difference in the control and treatment means by the standard deviation of the control group. I find that the effect sizes for all 5 variables with statistically significant differences had an effect size of less than 0.01 (Figure 3b); the d-statistics are robust to using the pooled standard deviation rather than the control group standard deviation. This indicates that they have a very small effect size: the control and treatment groups are balanced well.

5. Conclusion:

In conclusion, this report successfully replicates and extends the findings from Chakravorty et al (2023). The most important findings concern the importance of programs such as the DDU-SKY program for female employment and the need for targeted policies that consistently encourage women to remain in the labour force. Furthermore, a random forest model is used on the dataset to predict employment status and the number of jobs applied that participants for over the past two months in the final survey round, illustrating the importance of connections to obtain formal employment as opposed to formal methods such as online job portals. However, this should not be viewed in isolation: other strong predictors of employment status illustrate the merits of personality and consistent effort.

6. Appendix:

Figure 1a: Replication of Table 2

	Salaried Employment			
	(1)	(2)	(3)	(4)
Panel A: Survey Round 1 (Pre-lockdown 2020)				
Trained	0.309***	0.332***	0.322***	0.318***
Std error	(0.031)	(0.029)	(0.029)	(0.029)
Observations	2204	2204	2204	2204
Dropout mean	0.147	0.147	0.147	0.147
Panel B: Survey Round 1 (Jun-Jul 2020)				
Trained	0.235***	0.249***	0.242***	0.237***
Std error	(0.028)	(0.027)	(0.027)	(0.027)
Observations	2204	2204	2204	2204
Dropout mean	0.08	0.08	0.08	0.08
Panel C: Survey Round 2 (Mar-Apr 2021)				
Trained	0.123***	0.139***	0.132***	0.135***
Std error	(0.030)	(0.030)	(0.030)	(0.030)
Observations	1890	1890	1890	1890
Dropout mean	0.134	0.134	0.134	0.134
Panel D: Survey Round 3 (Nov-Dec 2021)				
Trained	0.139***	0.157***	0.161***	0.162***
Std error	(0.030)	(0.030)	(0.031)	(0.031)
Observations	1916	1916	1916	1916
Dropout mean	0.133	0.133	0.133	0.133

Figure 1b: Regression estimates including interactive variable between training and gender

	(1)	(2)	(3)	(4)
	Salaried Employment: Pre- lockdown 2020	Salaried Employment: Jun- Jul 2020	Salaried Employment: Mar- Apr 2021	Salaried Employment: Nov- Dec 2021
Trained	0.255*** (0.037)	0.175*** (0.035)	0.118*** (0.037)	0.164*** (0.038)
Trained * Female	0.169*** (0.060)	0.164*** (0.056)	0.048 (0.062)	-0.005 (0.064)
Female	-0.083 (0.062)	-0.079 (0.058)	-0.091 (0.064)	-0.060 (0.066)
Constant	-0.093 (0.140)	0.023 (0.131)	0.107 (0.141)	0.021 (0.144)
Observations	2204	2204	1890	1916

Standard errors in parentheses

* p<0.10 ** p<0.05 *** p<0.01

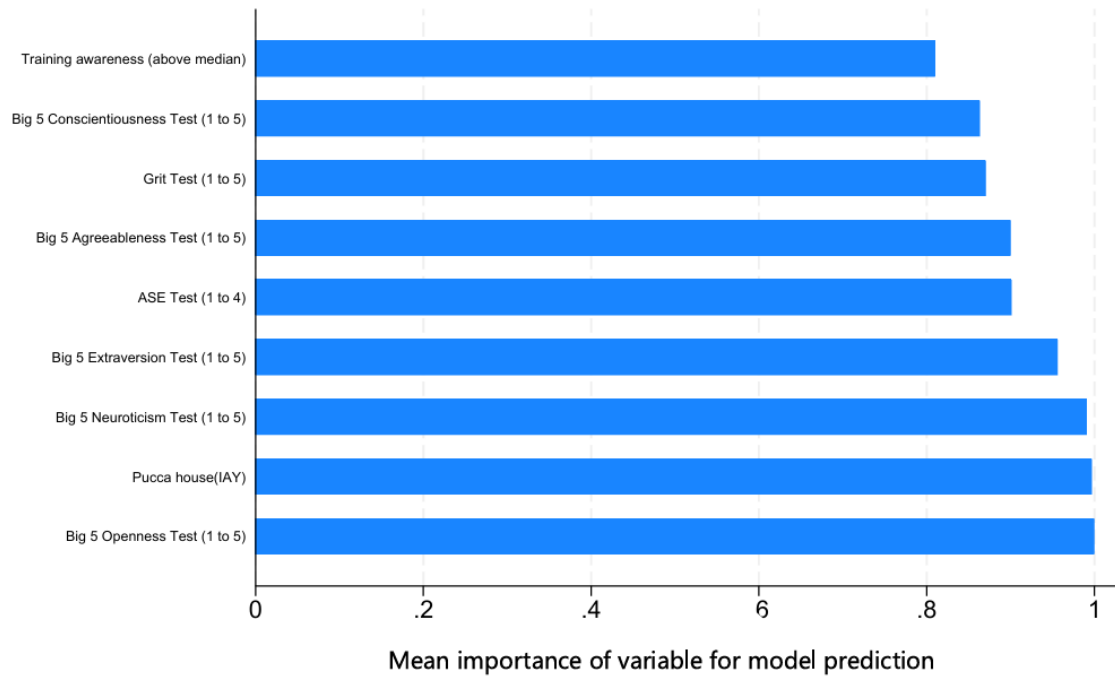
Figure 2a: Replication of Table 4, Panel B

	Applied for jobs in the last 2 months?	Number of job applications (1 - 2)	Number of job applications (3 or more)
Treatment	-0.011 (0.015)	-0.011 (0.013)	-0.004 (0.007)
P-value	0.416	0.388	0.573
Control mean	0.125	0.097	0.024
N	1955	1955	1955

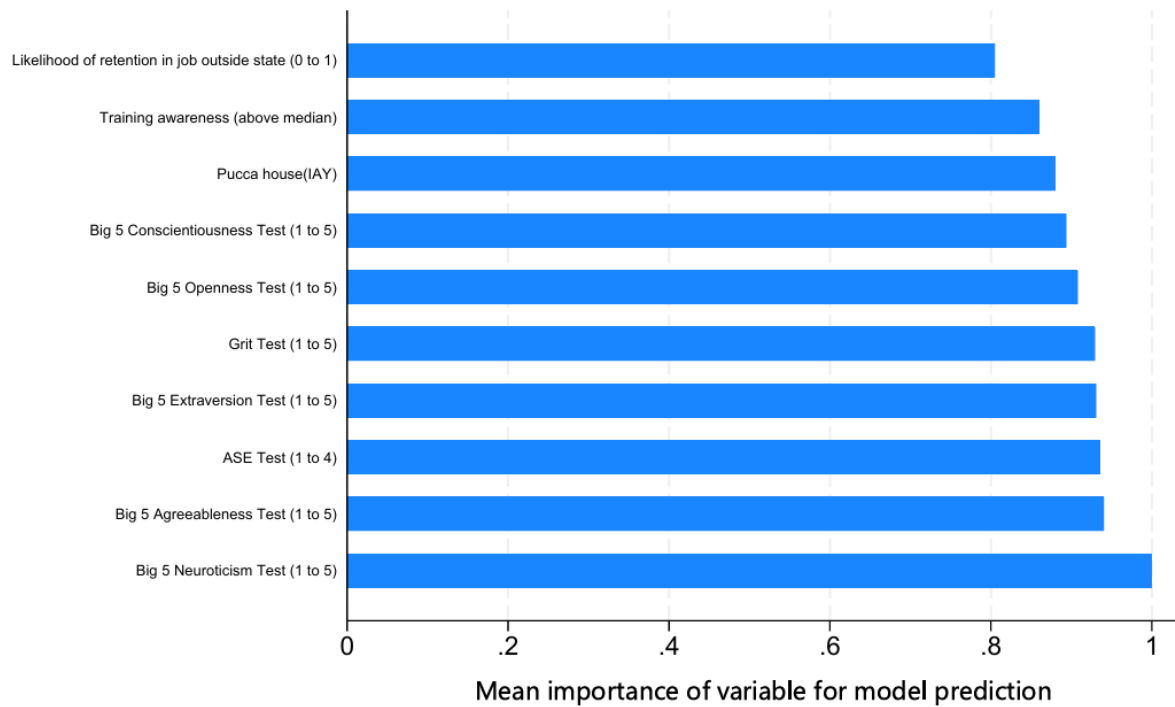
Standard errors in parentheses

Figure 2b: Most important variables for the Random Forest model in predicting each outcome

Most important variables for predicting Main Outcome 1 (Employment status):



Most important variables for predicting Main Outcome 2 (Having made 1 -2 job applications in the past 2 months):



Most important variables for predicting Main Outcome 3 (Having made 3 or more job applications in the past 2 months):

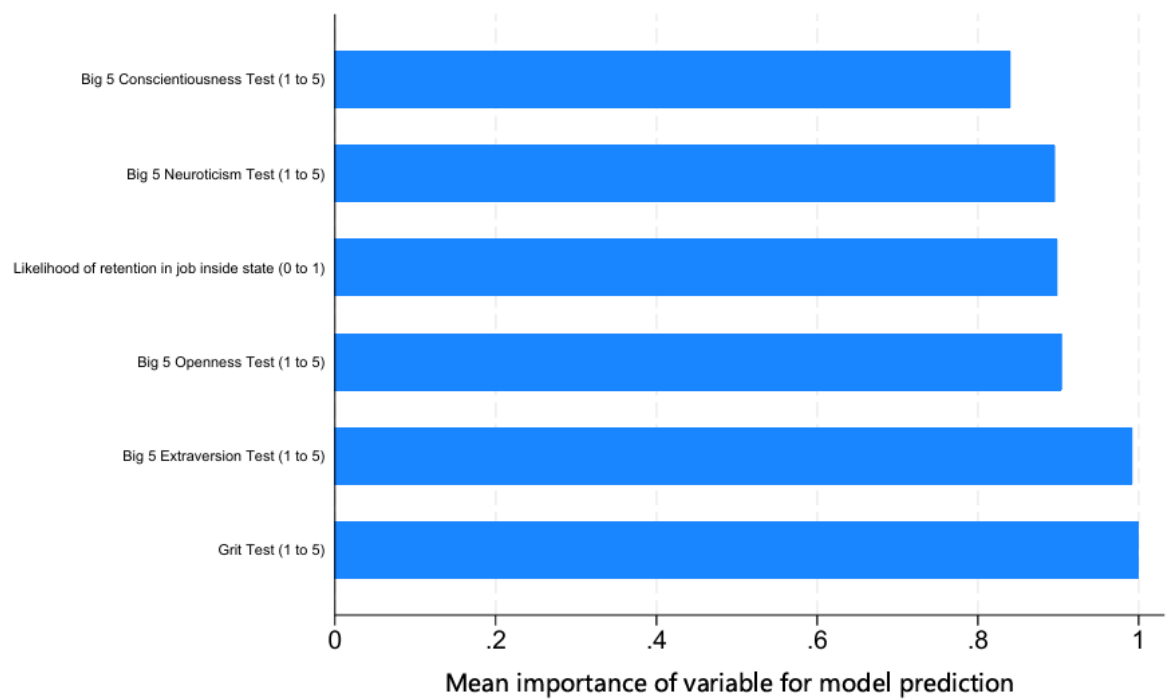


Figure 3a: Replication of Table A9

	(1) Control group	(2) Treatment group	(3) Difference (2-1)	(4) p-value
Panel A: Demographics and Caste				
Older(more than 20)	0.281	0.270	-0.010	0.578
Married	0.094	0.091	-0.004	0.756
Caste: ST	0.186	0.184	0.000	0.978
Caste: OBC	0.481	0.477	-0.005	0.790
Caste: General	0.073	0.062	-0.011	0.296
Religion: Muslim	0.062	0.057	-0.005	0.618
Religion: Christian	0.048	0.049	0.002	0.829
Panel B: Education				
Middle School (6-8 class)	0.045	0.053	0.008	0.394
Secondary level (9-10 class)	0.357	0.349	-0.008	0.702
Tertiary education (Graduate and above)	0.088	0.086	-0.002	0.840
Matric exam	0.933	0.936	0.003	0.798
More than 50% on the Matric exam	0.523	0.480	-0.044	0.037
Inter exam	0.583	0.585	0.002	0.940
Less than 50% on the Inter exam	0.237	0.230	-0.007	0.709
Panel C: Skills				
Big 5 Extraversion Test (1 - 5)	3.298	3.282	-0.016	0.485
Big 5 Agreeableness Test (1 - 5)	3.757	3.768	0.011	0.629
Big 5 Conscientiousness Test (1 - 5)	3.855	3.852	-0.003	0.907
Big 5 Neuroticism Test (1 - 5)	2.437	2.409	-0.027	0.354
Big 5 Openness Test (1 - 5)	3.945	3.923	-0.022	0.475
Grit Test (1 - 5)	3.408	3.429	0.021	0.432
ASE Test (1 - 4)	2.092	2.101	0.009	0.552
Life goal Test (1 -4)	2.136	2.151	0.015	0.296
Duration of baseline survey (above median)	0.505	0.496	-0.008	0.690
Panel D: Socioeconomic background				
Household head relationship (father)	0.097	0.083	-0.014	0.250
Household head relationship (mother)	0.069	0.086	0.017	0.131
Immediate difficulty to family	0.101	0.105	0.004	0.630
Future difficulty to family	0.140	0.150	0.010	0.339
Earning members (3 or more)	0.085	0.110	0.024	0.051
Household earning (15000 or more)	0.164	0.182	0.017	0.276
Household earning (5000 or less)	0.284	0.269	-0.015	0.428
Household earning (5001 - 9000)	0.230	0.226	-0.003	0.858
Agriculture land	0.660	0.651	-0.009	0.653
BPL card	0.797	0.781	-0.016	0.349
RSBY card	0.381	0.403	0.022	0.284
MNREGA	0.248	0.259	0.011	0.536
SHG number	0.739	0.737	-0.002	0.926
Semi pucca house	0.214	0.189	-0.025	0.142
Pucca house (IAY)	0.093	0.098	0.005	0.698
Pucca house (non IAY)	0.191	0.214	0.023	0.173

Own house	0.996	0.993	-0.003	0.391
Internet use	0.518	0.529	0.010	0.618
Joint household	0.058	0.078	0.019	0.065
Household members (2 or less)	0.059	0.054	-0.005	0.639
Household members (6 or more)	0.376	0.372	-0.005	0.815
Relatives migrated (1)	0.325	0.369	0.044	0.030
Relatives migrated (2 or more)	0.152	0.135	-0.018	0.234
Ever migrated out of state (relatives)	0.478	0.504	0.026	0.217
Ever migrated out of state (self)	0.120	0.139	0.019	0.169
Panel E: Expectations				
Previous earning	0.118	0.119	0.001	0.918
Hypothetical earning (immediate)	0.158	0.140	-0.019	0.197
Hypothetical earning (in one year)	0.232	0.225	-0.007	0.682
Expected earning (in one year)	0.406	0.386	-0.020	0.320
Preferred earning (in one year)	0.467	0.422	-0.045	0.028
Training awareness	0.532	0.542	0.010	0.401
Training usefulness	0.934	0.935	0.001	0.841
Training satisfaction	0.944	0.949	0.004	0.383
Likelihood of training completion	0.945	0.949	0.004	0.466
Likelihood of job offer	0.900	0.902	0.002	0.812
Expected minimum salary (immediate)	0.396	0.384	-0.013	0.502
Expected maximum salary (immediate)	0.409	0.408	-0.001	0.966
Expected average salary (immediate)	0.478	0.446	-0.033	0.110
Likelihood of accepting job offer outside state	0.786	0.798	0.011	0.224
Likelihood of accepting job outside state	0.824	0.829	0.005	0.587
Likelihood of accepting job inside state	0.841	0.836	-0.006	0.568
Likelihood of retention in job inside state	0.836	0.825	-0.011	0.265
Likelihood of retention in job outside state	0.818	0.818	0.001	0.949
Internet use	0.865	0.853	-0.012	0.395
Number of observations	1138	1122		

Figure 3b: Cohen's d coefficients:

Variable	Control mean	Treatment mean	p-value	Cohen's d
More than 50% on the Matric exam	0.523	0.48	0.037	0.0867
Earning members (3 or more)	0.085	0.11	0.051	-0.087
Joint household	0.058	0.078	0.065	-0.0833
Relatives migrated (1)	0.325	0.369	0.03	-0.0929
Preference earning (in one year)	0.467	0.422	0.028	0.0893

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